

Computational Tools for Multiscale Problems

Logistics

Course: CSCI 5526, Fall 2026

Time: Tuesday & Thursday 2:00–3:15 PM

Location: [University Club 6](#)

Instructor: Fruzsina Agocs

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Office hours: TBD

Course website: <https://cu-comptools.github.io/fall2026>

Course description

Physical phenomena are often described in terms of ordinary and partial differential equations (ODEs and PDEs) with interactions and features on multiple scales. Some of these systems are notoriously hard to solve but abound in scientific and engineering applications. In this course you will learn about state-of-the-art methods and software for the fast and accurate numerical solution of such systems, starting from their fundamental components (linear solve, interpolation, differentiation, quadrature) and building up to hybrid ODE solvers and boundary integral equation methods for PDEs.

Prerequisites

- Basic linear algebra, calculus, basic complex analysis
- CSCI 3656 (3) Numerical Computation encouraged
- CSCI 5636 Numerical Solution of Partial Differential Equations encouraged
- Familiarity with Python (or some other programming language) encouraged

Learning goals

- Be familiar with established and novel numerical tools (methods and software) for fundamental computational tasks: interpolation, quadrature, solution of ODEs, solution of linear PDEs with boundary integral equation methods.
- Understand how to assess whether a numerical solution is satisfactory, learn to evaluate what accuracy one can reasonably demand.
- Learn to build robust numerical code and (unit, convergence) test it.

Grading

Your grade will be calculated as follows:

- In-class/at-home hybrid assignments (90% total, equally weighed)
 - You will start assignments in class, working collaboratively with other students, and receiving guidance from the instructor.
 - You will have to finish the assignments at home and turn them in roughly a week's time.
 - There will be **roughly one of these every week - 2 weeks**.
 - The assignments will be graded and we will discuss common mistakes/conclusions in class.
 - Most assignments will include coding (in Python).
 - Use of AI/assistive coding is not allowed during these assignments (see the "Acceptable use of AI in this class" section below).
- Attendance (10% total)
 - This is a pass/fail criterion, you're required to attend at least 85% of classes (at least 24 out of 28).

Resources

- Trefethen, L. N. (2019). **Approximation theory and approximation practice**. (available [online](#) through author)
- Driscoll, T. A., & Braun, R. J. (2017). **Fundamentals of Numerical Computation**. (available [online](#) through author)

- Corless, R. M., & Fillion, N. (2013). **A graduate introduction to numerical methods**. ([PDF](#) through publisher)
- Hairer, E., Nørsett, S. P., & Wanner, G. **Solving Ordinary Differential Equations I-II**. ([PDF I](#) and [PDF II](#) through publisher)
- Trefethen, L. N. (2000). **Spectral methods in MATLAB**. (Chapter-by-chapter [PDFs](#) through publisher)
- Alex Barnett's Math 126 **Numerical analysis for PDEs and wave scattering** lecture notes (hand-written notes [online](#))
- Kress, R. (1999). **Linear integral equations**. ([PDF](#) through publisher)
- Further resources will be listed in the lecture notes

Course plan (subject to change)

- Fundamentals of computation
 - Floating point representation, condition number, backwards stability
- Components of differential equations
 - Interpolation, numerical differentiation, (smooth) quadrature
 - Complex contour deformation for oscillatory quadrature
- ODEs
 - Runge–Kutta methods, Butcher tableaus, stability and stiffness
 - spectral methods
 - asymptotic acceleration for linear oscillatory systems
 - Numerical Poincaré–Lindstedt for nonlinear oscillatory systems
- Integral equations I
 - What are integral equations? Nyström method, SVD, compact operators
- Integral equations II
 - Laplace's equation, fundamental solution, properties of harmonic functions, Green's representation theorem, boundary parameterization
- Integral equations III
 - Layer potentials, jump relations, interior/exterior Neumann/Dirichlet problems, complementary boundary value problem
- Integral equations IV

- Helmholtz equation, scattering formulation, combined field integral equation, transmission formulation, low-rank interactions
- Fast algorithms
 - The fast multipole method, iterative methods and their convergence, singular / near-singular quadrature

Acceptable use of AI in this class

You may not, under any circumstances, use LLMs or other ML-based tools (e.g. ChatGPT, GitHub copilot, Cursor) for assistance in your assignments. Doing so will count as cheating and will be reported, and you will get a zero on your assignment.

Classroom Behavior

Students and faculty are responsible for maintaining an appropriate learning environment in all instructional settings, whether in person, remote, or online. Failure to adhere to such behavioral standards may be subject to discipline. Professional courtesy and sensitivity are especially important with respect to individuals and topics dealing with race, color, national origin, sex, pregnancy, age, disability, creed, religion, sexual orientation, gender identity, gender expression, veteran status, marital status, political affiliation, or political philosophy.

Additional classroom behavior information

- [Student Classroom and Course-Related Behavior Policy](#).
- [Student Code of Conduct](#).
- [Office of Institutional Equity and Compliance](#).

Accommodation for Disabilities, Temporary Medical Conditions, and Medical Isolation

If you qualify for accommodations because of a disability, please submit your accommodation letter from Disability Services to your faculty member in a timely

manner so that your needs can be addressed. Disability Services determines accommodations based on documented disabilities in the academic environment. Information on requesting accommodations is located on the [Disability Services website](#). Contact Disability Services at 303-492-8671 or DSinfo@colorado.edu for further assistance. If you have a temporary medical condition, see [Temporary Medical Conditions](#) on the Disability Services website.

If you have a temporary illness, injury or required medical isolation for which you require adjustment, please contact the instructor via fruzsina.agocs@colorado.edu (you are **not** required to state the nature of your illness/injury or provide a doctor's note).

Student Names and Pronouns

CU Boulder recognizes that students' legal information does not always align with how they identify. If you wish to have a name other than your legal name appear on your instructors' class rosters and in Canvas, or if you wish to choose pronouns to appear on your instructors' class rosters and in Canvas, visit the [Registrar's website](#) for instructions on how to change your personal information in university systems.

Honor Code

All students enrolled in a University of Colorado Boulder course are responsible for knowing and adhering to the [Honor Code](#). Violations of the Honor Code may include but are not limited to: plagiarism (including use of paper writing services or technology [such as essay bots]), cheating, fabrication, lying, bribery, threat, unauthorized access to academic materials, clicker fraud, submitting the same or similar work in more than one course without permission from all course instructors involved, and aiding academic dishonesty. Understanding the course's syllabus is a vital part of adhering to the Honor Code.

All incidents of academic misconduct will be reported to Student Conduct & Conflict Resolution: StudentConduct@colorado.edu. Students found responsible for violating the Honor Code will be assigned resolution outcomes from Student Conduct &

Conflict Resolution and will be subject to academic sanctions from the faculty member. Visit [Honor Code](#) for more information on the academic integrity policy.

Sexual Misconduct, Discrimination, Harassment and/or Related Retaliation

CU Boulder is committed to fostering a productive and welcoming learning, working, and living environment. University policy prohibits [protected-class](#) discrimination and harassment, sexual misconduct (harassment, exploitation, and assault), intimate partner abuse (dating or domestic violence), stalking, and related retaliation by or against members of our community on- or off-campus. Denial of an approved accommodation for disability, religious observance, or pregnancy or pregnancy related medical conditions may be discriminatory.

The [Office of Institutional Equity and Compliance \(OIEC\)](#) addresses these concerns, and individuals who have been subjected to misconduct can contact OIEC at 303-492-2127 or email OIEC@colorado.edu. Information about university policies, [OIEC reporting options](#), and [OIEC support resources](#) including confidential services can be found on the [OIEC website](#).

Faculty and graduate instructors are required to inform OIEC when someone discloses misconduct regardless of when or where it occurred. This is to ensure that those impacted receive outreach from OIEC about resolution options and support resources. To learn more about reporting and support options for a variety of concerns, visit the [Don't Ignore It page](#).

Accommodation for Religious Obligations

Instructional faculty members must make every reasonable effort to accommodate all students who have conflicts with scheduled exams, assignment deadlines or required attendance due to a religious observance. Whenever possible, students must notify the instructional faculty member at least two weeks in advance of the expected exam, assignment deadline, or attendance conflict to request an

accommodation for religious observance. If the start date of the course is less than two weeks before the date of requested accommodation, students must notify the instructional faculty member on the start date of the course. See the [Student Academic Accommodations for Religious Observances Policy](#) for more information.

In this class, please inform the instructor via fruzsina.agocs@colorado.edu ahead of time if you need religious accommodation.

Mental Health and Wellness

The University of Colorado Boulder is committed to the well-being of all students. If you are struggling with personal stressors, mental health or substance use concerns that are impacting academic or daily life, please contact [Counseling and Psychiatric Services \(CAPS\)](#) located in C4C or call (303) 492-2277, 24/7.